NEET
SAMPLE PAPER 1- 2020

PHYSICS
QUESTION PAPER

TRIPATHI STUDY ZONE
JEE MAIN | NEET | AIIMS | MHCET | BITSAT
Physics Expert
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1. A point charge q is placed at a distance a / 2 directly above the centre of a square of side a. The electric flux through the square is :-

(A) $\frac{q}{\epsilon_0}$ (B) $\frac{q}{\pi \epsilon_0}$ (C) $\frac{q}{4 \epsilon_0}$ (D) $\frac{q}{6 \epsilon_0}$

2. A heavy body moving with a velocity 30 m/ s and another small object at rest undergo an elastic collision. The latter will move with a velocity of :-

(A) 30 m/s (B) 60 m/s (C) 80 m/s (D) zero

3. A vertical mass spring system executes simple harmonic oscillations with a period of 2s. A quantity of this system which exhibits simple harmonic variation with a period of 1 sec is :-

(A) Velocity

(B) Potential energy

(C) Phase difference between acceleration and displacement

(D) Difference between kinetic energy and potential energy

4. If a photocell is illuminated with a radiation of 1240 Å, the stopping potential is found to be 8 volt; then the work function of the emitter and the threshold wavelength are :-

(A) 2 eV, 2000 Å (B) 2 eV, 6200 Å (C) 2 eV, 2480 Å (D) 3 eV, 6200 Å

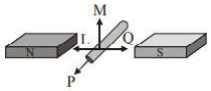
5. A thin convex lens of focal length 5 cm is used as a simple microscope by a person with normal near point located (25 cm) away. What is the magnifying power of the microscope ?

(A) 5 (B) 6 (C) 10 (D) 12

6. An atomic nucleus $_{90}Th^{232}$ emits several α and β-radiations and finally reduces to $_{82}Pb^{208}$. It must have emitted :-

(A) 4α and 2β (B) 6α and 4β (C) 8α and 24B (D) 4α and 16β

7. An electric potential difference will be induced between ends of the conductor shown in the diagram, when the conductor moves in the direction :-



(A) P (B) Q (C) L (D) M

8. With rise of temperature the surface tension of a liquid :-

(A) decreases (B) increases (C) remains constant (D) none of these

9. A stream-lined body falls through air from a height h on the surface of liquid. Let d and D denote the densities of the materials of the body and the liquid respectively. If D > d, then the time after which the body will be instantaneously at rest, is :-

(A) $\sqrt{\frac{2h}{g}}$ (B) $\sqrt{\frac{2h}{g}\frac{D}{d}}$ (C) $\sqrt{\frac{2h}{g}\frac{d}{D}}$ (D) $\sqrt{\frac{2h}{g}\left(\frac{d}{D-d}\right)}$

10. A pipe 17 cm long is closed at one end. Which harmonic mode of the pipe resonates a 1.5 kHz source ? (Speed of sound in air = 340 m s^{-1}) :-

(A) First (B) Third (C) Fifth (D) Seventh

11. $\vec{A}$ and $\vec{B}$ are two vectors given by $\vec{A} = 2\hat{i} + 3\hat{j}$ and $\vec{B} = \hat{i} + \hat{j}$ The magnitude of the component (projection) of $\vec{A}$ on $\vec{B}$ is :

(A) $5/\sqrt{2}$ (B) $3/\sqrt{2}$ (C) $7/\sqrt{2}$ (D) $1/\sqrt{2}$

12. The electric field of a plane electromagnetic wave varies with time of amplitude 2 Vm^{-1} propagating along z-axis. The average energy density of the magnetic field (in J m–3) is :-

(A) 13.29×10^{-12} (B) 8.86×10^{-12} (C) 17.72×10^{-12} (D) 4.43×10^{-12}

13. A parallel plate air capacitor has a capacitance C. When it is half filled with a dielectric of dielectric constant 5, the percentage increase in the capacitance will be :-

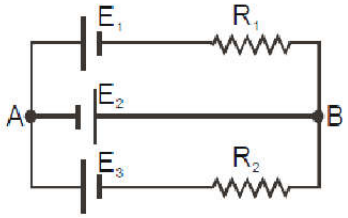


(A) 400% (B) 66.6% (C) 33.3% (D) 200%

14. What is the value of linear velocity if $\vec{r} = 3\hat{i} + 4\hat{j} + 6\hat{k}$ and $\vec{\omega} = -5\hat{i} + 3\hat{j} + 5\hat{k}$

(A) $-2\hat{i} + 45\hat{j} - 29\hat{k}$ (B) $2\hat{i} - 45\hat{j} + 29\hat{k}$ (C) $3\hat{i} - 29\hat{j} + 45\hat{k}$ (D) $5\hat{i} - 6\hat{j} + 45\hat{k}$

15. In the circuit shown here, $E_1 = E_2 = E_3 = 2$ V and $R_1 = R_2 = 4$ ohms. The current flowing between points A and B through battery E_2 is:



(A) zero (B) 2 A from A to B (C) 2 A from B to A (D) none of the above

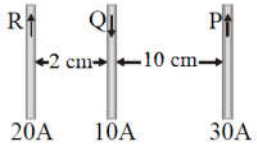
16. A body of mass 2.9 kg is suspended from a string of length 2.5 m and is at rest. A bullet of mass 100 g , strikes the block horizontally with velocity 150 ms^{-1} and sticks to it. What is the maximum angle made by the string with the vertical after the impact ? ($g = 10$ m/s^2)

(A) 30° (B) 45° (C) 60° (D) 90°

17. Pressure exerted by a gas is found to be 50 N/m^2 , then the value of pressure in CGS system is :-

(A) 0.5 (B) 0.05 (C) 5000 (D) 500

18. Three long, straight and parallel wires carrying currents are arranged as shown in figure. The force experienced by 10 cm length of wire Q is:-



(A) 1.4×10^{-4} N towards right (B) 1.4×10^{-4} N towards left

(C) 2.6×10^{-4} N towards right (D) 2.6×10^{-4} N towards left

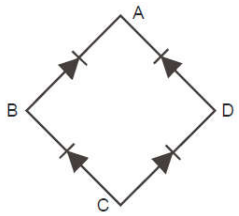
19. The magnetic field at the centre of an equilateral triangular loop of side 2L and carrying a current i is:-

(A) $\frac{9\mu_0 i}{4\pi L}$ (B) $\frac{3\sqrt{3}\mu_0 i}{4\pi L}$ (C) $\frac{2\sqrt{3}\mu_0 i}{\pi L}$ (D) $\frac{3\mu_0 i}{4\pi L}$

20. Three identical bodies of equal mass M each are moving along a circle of radius R under the action of their mutual gravitational attraction. The speed of each body is :-

(A) $\sqrt{\frac{GM}{R}}$ (B) $\sqrt{\frac{GM}{3R}}$ (C) $\sqrt{\frac{GM}{\sqrt{3}R}}$ (D) $\sqrt{\frac{GM}{\sqrt{2}R}}$

21. For the given circuit shown in fig, to act as full wave rectifier :– a.c. input should be connected across ..... and ..... the d.c. output would appear across ..... and .....



(A) A, C, B, D (B) B, D, A, C (C) A, B, C, D (D) C, A, D, B

22. A hot liquid kept in a beaker cools from 80°C to 70°C in 2 min. If the surrounding temperature is 30°C, then the time of cooling of the same liquid from 60°C to 50°C :-

(A) 240 s (B) 360 s (C) 480 s (D) 216 s

23. Correct statement for diode is :–

(A) in full wave rectifier both diodes work alternatively.

(B) in full wave rectifier both diodes work simultaneously.

(C) efficiency of full wave rectifier and half wave rectifier is same.

(D) full wave rectifier in bidirectional.

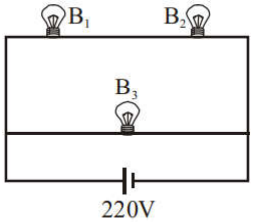
24. A body falling on the ground from a height of 10 m, rebounds to a height 2.5 m, then the ratio of the velocities of the body just before and after the collision will be :-

(A) 2 (B) 4 (C) 0.5 (D) 0.25

25. A particle is performing simple harmonic motion along x-axis with amplitude 4 cm and time period 1.2 sec. The minimum time taken by the particle to move from x = 2 cm to x = + 4 cm and back again is given by :-

(A) 0.6 s (B) 0.4 s (C) 0.3 s (D) 0.2 s

26. A 100 W bulb B_1 and two 60 W bulbs B_2 and B_3, are connected to a 220 V source, as shown in Figure. Now P_1, P_2 and P_3 are the output powers of the bulbs B_1, B_2 and B_3 respectively. Then:

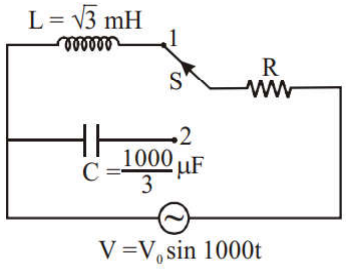


(A) $P_1 > P_2 = P_3$ (B) $P_1 > P_2 > P_3$ (C) $P_1 < P_2 = P_3$ (D) $P_1 < P_2 < P_3$

27. An inductor (L)-resistance (R), battery circuit is switched on at t = 0. If the e.m.f. of battery is E, the charge passing through the battery in time constant T is :-

(A) $\frac{ET}{Re}$ (B) $\frac{eET}{R}$ (C) $\frac{ET}{R\pi e}$ (D) $\frac{2ET}{R\,e}$

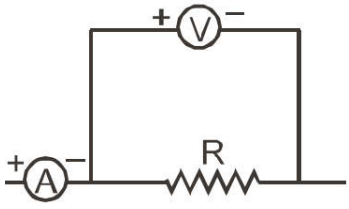
28. In the given AC circuit, when switch S is at position 1, the source emf leads current by π/6. Now, if the switch is at position 2, then :-



(A) current leads source emf by $\frac{\pi}{4}$ (B) current leads source emf by $\frac{\pi}{3}$

(C) source emf lead current by $\frac{\pi}{4}$ (D) source emf leads current by $\frac{\pi}{3}$

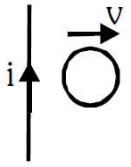
29. A student connects a voltmeter, ammeter and resistance according to the circuit given. If the voltmeter reading is 20 V and ammeter reading is 4 A, then the resistance will be



(A) equal to 5 Ω (B) more than 5 Ω

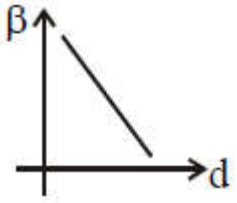
(C) less than 5 Ω (4) less or more depending on the material of wire

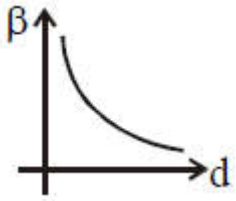
30. A circular loop of radius r is moved away from a current carrying wire then induced current in circular loop will be :-

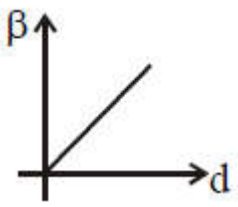


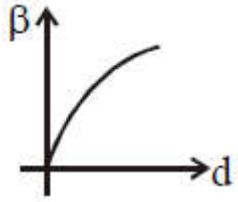
(A) Clock wise (B) Anti clockwise (C) Not induced (D) None of them

31. Correct curve between fringe width β and distance between slits(d) is :-

(A) 

(B) 

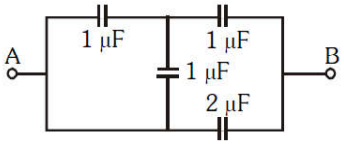
(C) 

(D) 

32. A flywheel is making $3000/\pi$ revolutions per minute about its axis. If the moment of inertia of the flywheel about that axis is 400 kgm^2, its rotational kinetic energy is

(A) 2×10^6 J (B) 3×10^3 J (C) $500\pi^2$J (D) 12×10^3 J

33. The equivalent capacitance between points A and B of the circuit shown will be :



(A) 2/3 μF (B) 5/3 μF (C) 8/3 μF (D) 7/3 μF

34. If ratio of maximum and minimum intensities is 36 : 1 in an interference pattern then, ratio of amplitudes of interfering waves is :-

(A) 3 : 7 (B) 7 : 4 (C) 4 : 7 (D) 7 : 5

35. What is the angle between the electric dipole moment and the electric field strength due to it on the equatorial line :-

(A) 0° (B) 90° (C) 180° (D) 45°

36. The electric potential at any point as a function of distance (x) in meter is given by $V = 5x^2 + 10x - 9$ (volt). Value of electric field at x = 1 is :-

(A) -20 Vm^{-1} (B) 6 Vm^{-1} (C) 11 Vm^{-1} (D) -23 Vm^{-1}

37. 2g of steam condenses when passed through 40g of water initially at 25°C. The condensation of steam raises the temperature of water to 54.3°C, what is the latent heat of steam?

(A) 540 cal/g (B) 536 cal/g (C) 270 cal/g (D) 480 cal/g

38. Two bodies are thrown up at angles of 45° and 60°, respectively, with the horizontal. If both bodies attain same vertical height, then the ratio of velocities with which these are thrown is :-

(A) $\sqrt{\frac{2}{3}}$ (B) $\frac{2}{\sqrt{3}}$ (C) $\sqrt{\frac{3}{2}}$ (D) $\frac{\sqrt{3}}{2}$

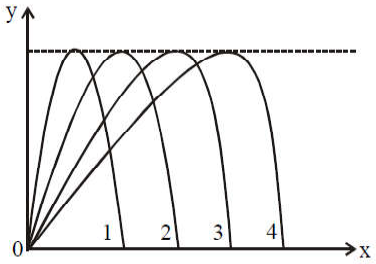
39. Ten one–rupeed coins are put on top of each other on a table. Each coin has mass m. The reaction of the 6th coin (counted from the bottom) on the 7th coins is

(A) 4mg (B) 6mg (C) 7mg (D) 3mg

40. What should be the maximum acceptance angle at the air-core interface of an optical fibre if n_1 and n_2 are the refractive indices of the core and the cladding respectively :-

(A) $\sin^{-1}(n_2/n_1)$ (B) $\sin^{-1}\sqrt{n_1^2 - n_2^2}$ (C) $\tan^{-1}\left(\frac{n_2}{n_1}\right)$ (D) $\tan^{-1}\left(\frac{n_1}{n_2}\right)$

41. Figure shows four paths for a kicked football. lgnoring the effects of air on the flight, rank the paths according to initial horizontal velocity component, highest first :-

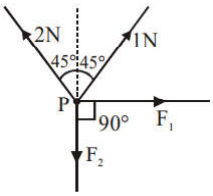


(A) 1,2,3,4 (B) 2,3,4,1 (C) 3,4,1,2 (D) 4,3,2,1

42. A block of mass m rests on a rough inclined plane. The coefficient of friction between the surface and the block is μ. At what angle of inclination θ of the plane to the horizontal will the block just start to slide down the plane?

(A) $\theta = \tan^{-1}\mu$ (B) $\theta = \cos^{-1}\mu$ (C) $\theta = \sin^{-1}\mu$ (D) $\theta = \sec^{-1}\mu$

43. There are four force acting at a point P produced by strings as shown in figure, which is at rest. The forces F_1 and F_2 are



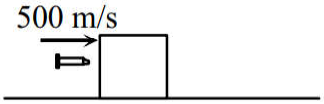
(A) $\frac{1}{\sqrt{2}}N, \frac{3}{\sqrt{2}}N$ (B) $\frac{3}{\sqrt{2}}N, \frac{1}{\sqrt{2}}N$ (C) $\frac{1}{\sqrt{2}}N, \frac{1}{\sqrt{2}}N$ (D) $\frac{3}{\sqrt{2}}N, \frac{3}{\sqrt{2}}N$

44. For waves propagating in a medium, identify the property that is independent of the others:-

(A) Velocity (B) Wavelength

(C) Frequency (D) All these depend on each other

45. A bullet of mass 20 g travelling horizontally with a speed of 500 m/s passes through a wooden block of mass 10.0 kg initially at rest on a surface. The bullet emerges with a speed of 100 m/s and the block slides 20 cm on the surface before coming to rest, the coefficient of friction between the block and the surface. ($g = 10\ m/s^2$)



(A) 0.16 (B) 0.6 (C) 0.5 (D) 0.25

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